

CT UNIVERSITY
END SEMESTER EXAMINATION

Subject: GA

Time Allowed: 3 Hours | Maximum Marks: 70 | 25 May 2026

Name: _____ Roll No.: _____
Class/Section: _____

GENERAL INSTRUCTIONS

1. Answer all questions.
2. Part A is compulsory.
3. In Part B, answer any 4 out of 6.
4. In Part C, answer any 2 out of 3.

PART A — OBJECTIVE (COMPULSORY)

Answer all 10 questions. 1 mark each.

Q1. (1 marks)

What happens when a fault occurs in the electrical system?

- (a) The MCB will trip and interrupt the current flow
- (b) A fuse will melt, breaking the circuit

Study the figure below.

Q2. (1 marks)

Which statement about a concave lens' ability to form images is true?

Q3. (1 marks)

If an object is placed at twice the focal length from a concave mirror, what can be said about the image formed?

- (a) The image will also be real and inverted
- (b) No image will form

Q4. (1 marks)

A terminal is the point at which two conductors are connected to an electrical circuit. Which of these options best describes a device that connects multiple cells in series or parallel?

- (a) Battery
- (b) Resistor
- (c) Capacitor
- (d) Diode

Study the figure below.

Q5. (1 marks)

When current flows through a room heater, it encounters resistance. This results in the generation of heat to warm up the air inside the space around the appliance. Which device is specifically designed for this purpose?

- (a) Resistor
- (b) Transformer
- (c) Thermostat
- (d) Fuse

Q6. (1 marks)

A fuse is a safety device that protects an electrical circuit from the damage caused by excessive current, which can result in overheating. Which of these statements about a fuse are true?

- (a) T
- (b) F

Study the figure below.

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Q7. (1 marks)

Which statement accurately describes the working model of a railway signal as represented in one of these figures?

- (a) The figure shows that its proportion in air is about 3%.
- (b) The correct option cannot be determined without referring to specific figures.
- (c) All four statements are accurate descriptions of the railway signal model.
- (d) Only two options accurately describe aspects related to fuses and MCB.

Study the figure below.

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Q8. (1 marks)

Explain an important concept from the syllabus related to A in the context of the uploaded material. (1 marks)

PART B — SHORT ANSWER

Answer any 4 of the following 6 questions. 5 marks each.

Q9. (5 marks)

Refer to the provided speed comparison context. Based on your understanding of motion and velocity from this section, which one would you say is moving faster between two vehicles A and B?

Q10. (5 marks)

. Distance-Time Graph for the Motion of Two Vehicles A and B is shown in this figure. Which one shows a truck moving with speed which isn't constant?

Figure

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Q11. (5 marks)

Refer to the provided context on speed and motion. If you were asked about which activity has a slower or faster rate of breathing based on your experience, how would you assign numbers in order of increasing breathing rates?

Q12. (5 marks)

(a) Which parts of the black sheep have wool?

Q13. (5 marks)

(b) What is the correct term for a young caterpillar?

PART C — LONG ANSWER

Answer any 2 of the following 3 questions. 10 marks each.

Q14. (10 marks)

Given that it takes 32 seconds for the simple pendulum in an experiment to complete 20 full oscillations (from one extreme end of its swing back to the other), calculate the time period 'T' of a single oscillation.

Q15. (10 marks)

If the railway track connects City X to City Y with a total length of 240 kilometers and the train completes this journey in exactly 4 hours without any stops or delays, what is the constant speed 'v' at which the train travels?